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  `https://github.com/julianmak/OCES5303_ML_ocean`

  The repository principally contains the compiled products rather than the source for size reasons.
- Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 5303 :
ML methods in Ocean Sciences

Session 10: FNOs

Outline

- Fourier Neural Operators

  → the keyword is the **operator** part

  → quick recap/overview of Fourier analysis

  → structure of FNO

- FNO for three examples

  → 1d FNO for linear PDE

  → 1d FNO for nonlinear PDE

  → 2d FNO for image regression

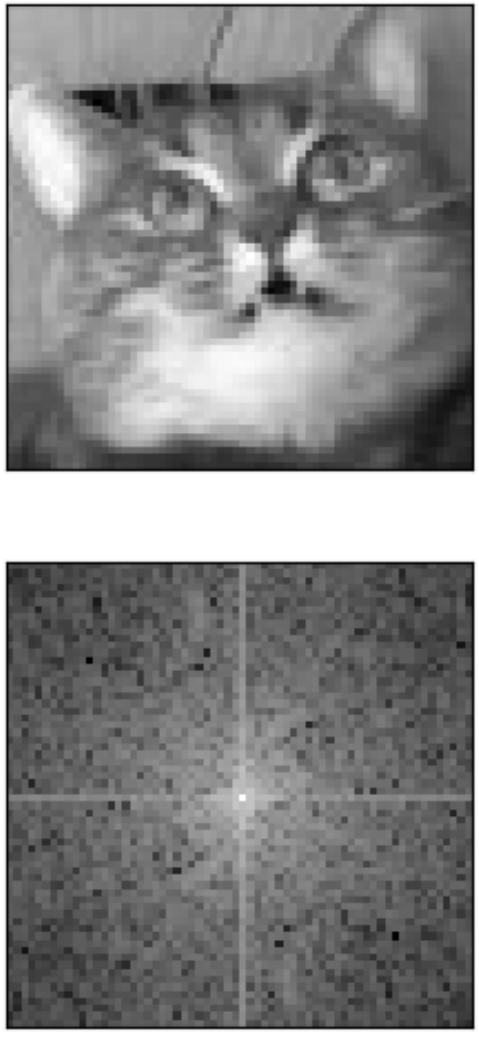

Figure: Head ad-hoc TA and her **power spectrum**.

Neural Networks vs. Neural **Operators**

- **neural networks** are intended to be maps between vectors that live in some **finite** dimensional data space

  $\rightarrow$ classification of cats/dogs: (64^2)d vector to 1d number

  $\rightarrow$ dimension reduction of cats/dogs: (64^2)d to 4d say

  $\rightarrow$ regression of cats/dogs: (64×32)d vector to (64×32)d vector

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- **operators** map between **infinite** dimension spaces

  $\rightarrow$ idea: learn the mapping between **spaces**, generalise better to unseen data?

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- in practice we cannot deal with infinite dimension spaces directly

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  $\rightarrow$ idea: take a "good" finite representation of the underlying space, and learn operations acting on that representation

- Fourier representation is one of these

  $\rightarrow$ basically **sines** and **cosines**

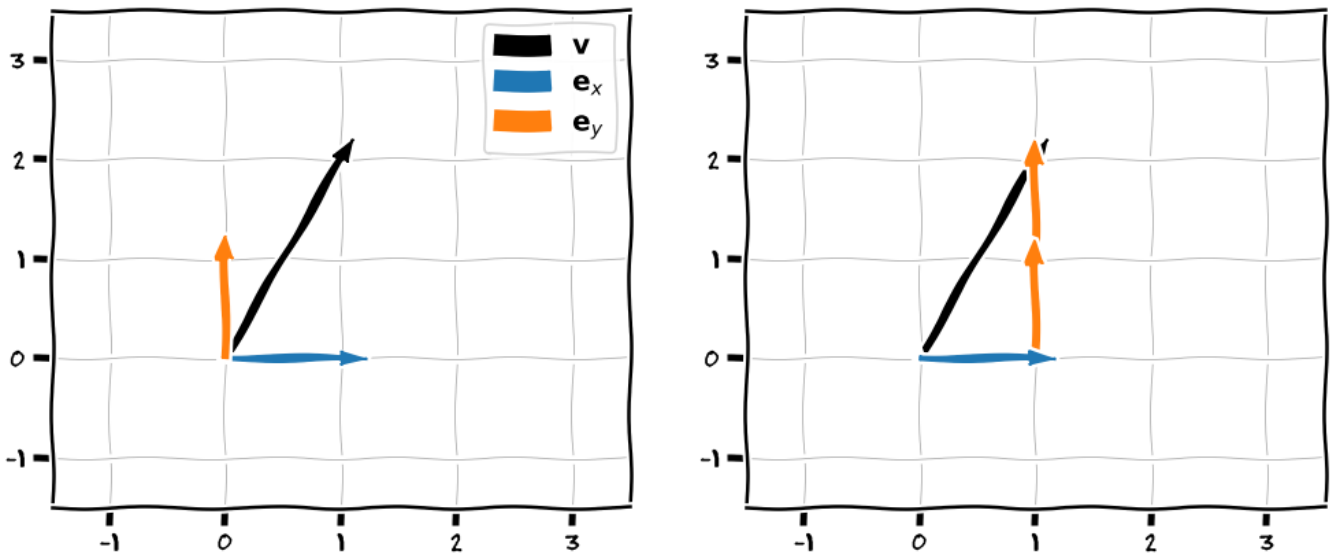
  $\rightarrow$ can represent a wide class of functions naturally or by artificial modifications

  $\rightarrow$ others entirely possible (e.g. spherical harmonics)

Recap: Fourier analysis

- standard basis in $\mathbb{R}^2$ is $e_1 = (1, 0)$ and $e_2 = (0, 1)$, so e.g.

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = e_1 + 2e_2$$



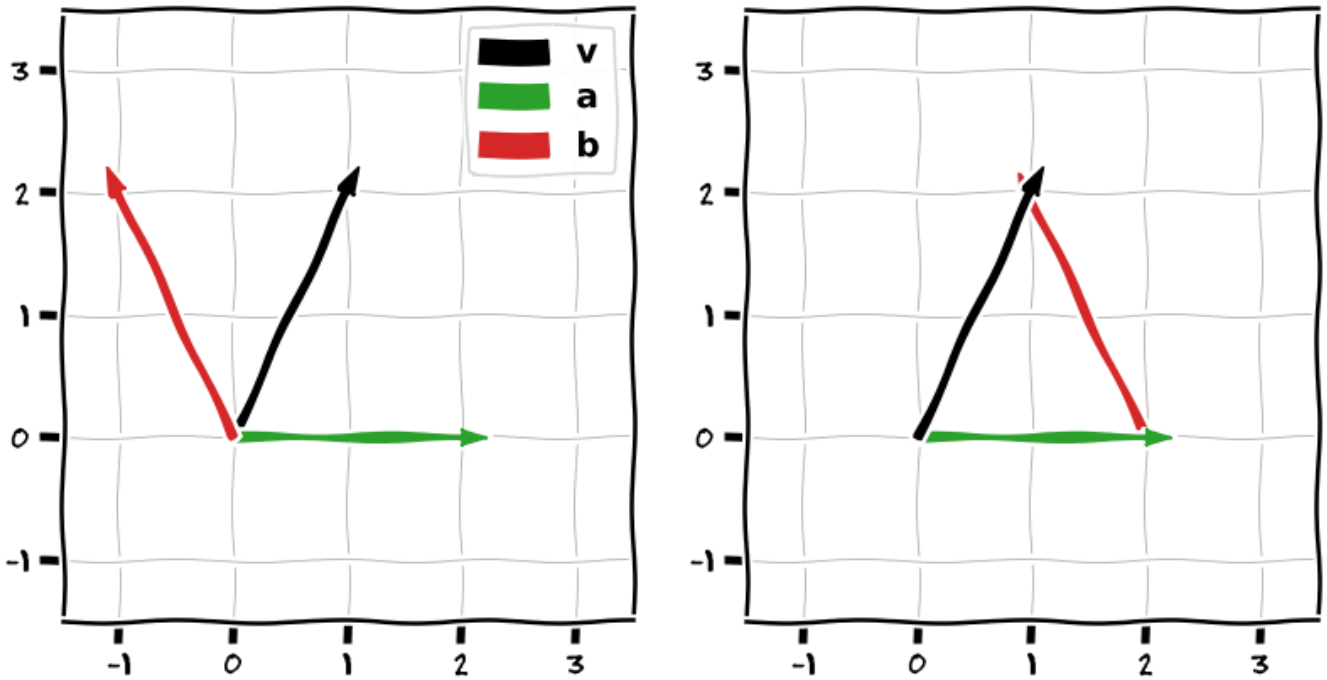
- $\{e_1, e_2\}$ form a **basis** that happens to be **orthogonal**

(with respect to the usual metric; actually ortho**normal** here)

Recap: Fourier analysis

- could do instead $\boldsymbol{a} = (2, 0)$ and $\boldsymbol{b} = (-1, 2)$, so e.g.

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \boldsymbol{a} + \boldsymbol{b}$$



- $\{\boldsymbol{a}, \boldsymbol{b}\}$ is still a basis but not orthogonal
  $\rightarrow$ representation of $\boldsymbol{v}$ in this basis would be $(1, 1)$

Recap: Fourier analysis

- idea of a basis is quite general, e.g. color = (R, G, B)
  $\rightarrow$ cf. lego pieces for the larger structure

Recap: Fourier analysis

- idea of a basis is quite general, e.g. color = (R, G, B)

  $\rightarrow$ cf. lego pieces for the larger structure

- orthogonality is useful because then we have minimal overlaps between the co-ordinates/channels

  $\rightarrow$ cf. non-confounding variables

  $\rightarrow$ PCA/EOF looks for an orthogonal representation from data (the O in EOF)

would like an orthogonal basis for functions

Recap: Fourier analysis

- TL;DR: sines and cosines form an orthogonal basis for a wide class of functions, i.e.

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx) = \text{Real}\left(\sum_{k=-\infty}^{\infty} c_k e^{ikx}\right)$$

- f has a **real** space representation of (for $f(x_i) = f_i$)

$$f(x) = [f_0, f_1, \ldots f_N]$$

and a Fourier/**spectral** representation of (imprecise with $N//2$ below)

$$\hat{f}(k) = [c_0, c_1, \ldots c_{\lfloor N/2 \rfloor}, c_{-\lfloor N/2 \rfloor}, \ldots c_{-1}]$$

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- really for L^2 periodic functions
  $\rightarrow$ can manipulate functions to force them to be L^2 periodic

(e.g. padding, tapering)

Recap: Fourier analysis

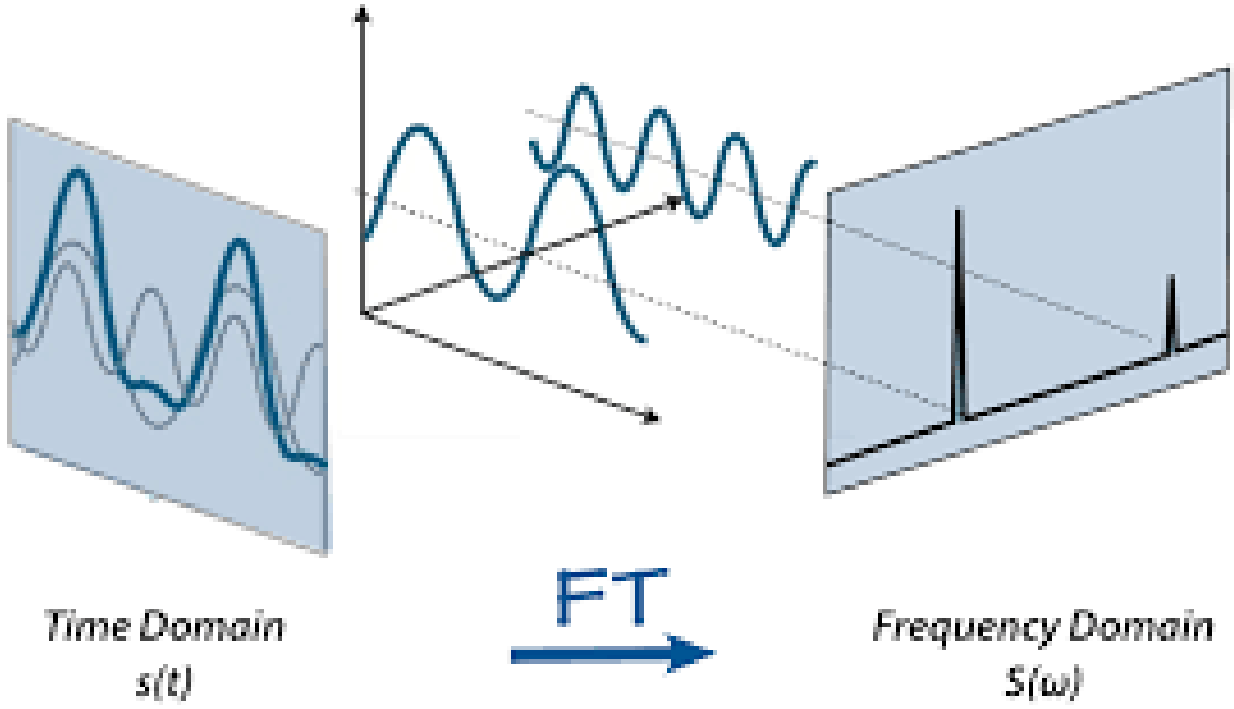

Figure: Schematic for the idea behind Fourier or spectral representation. From the internet (not sure where to attribute...)

- Fourier transform from real to spectral representation
  → e.g. signal analysis, imaging, etc.

Recap: Fourier analysis

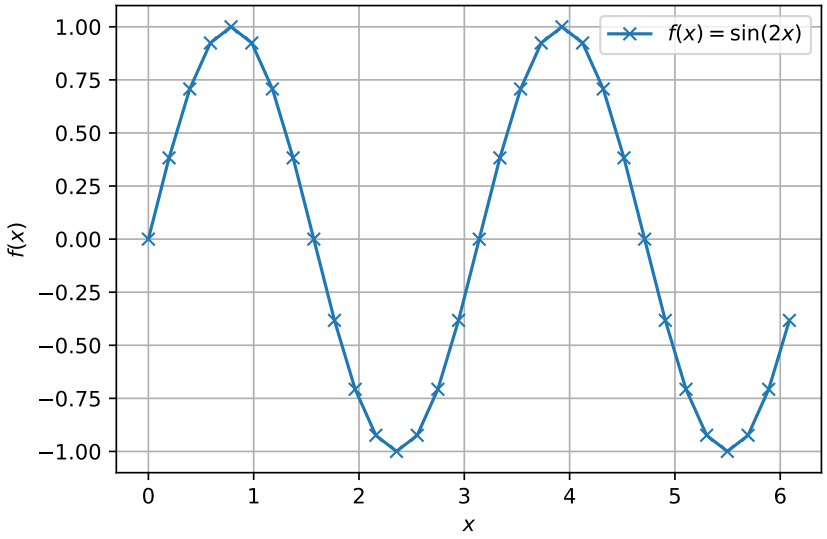

Figure: Example sine wave.

- take domain to be $[0, 2\pi]$ and $f(x) = \sin(2x)$

Q. what should the spectral representation look like?

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx) = \text{Real}\left(\sum_{k=-\infty}^{\infty} c_k e^{ikx}\right)$$

Recap: Fourier analysis

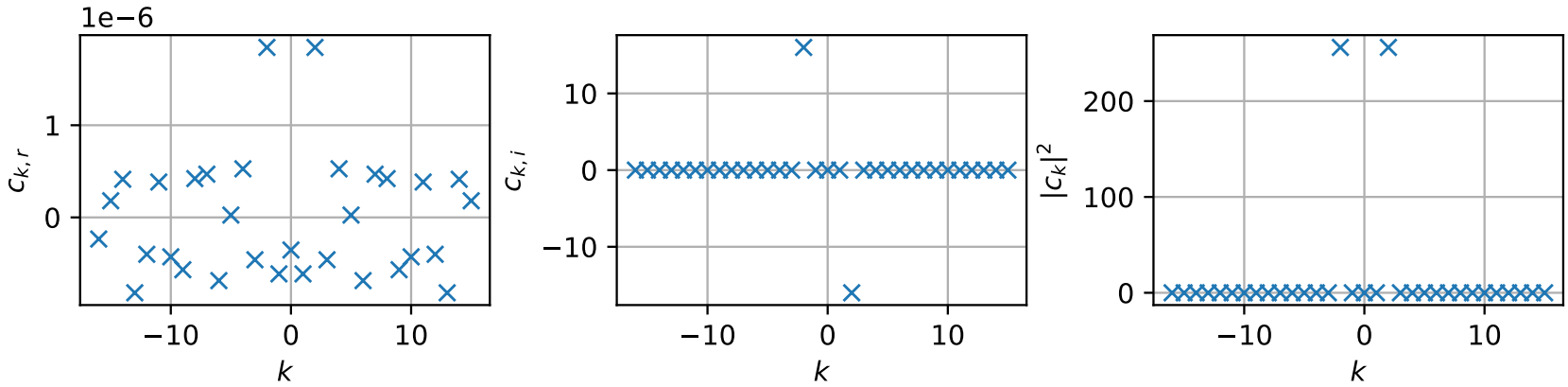

Figure: Fourier transform of the sine wave. (*a*) Real and (*b*) imaginary part of the Fourier coefficients, and (*c*) the power spectrum.

- done through `torch` (could use `numpy` too)

  $\rightarrow$ real components of c_k are tiny

  $\rightarrow$ non-zero imaginary components of c_k only at $k = 2$

  $\rightarrow$ non-zero power spectrum (i.e. $|c_k|^2$) only at $k = 2$

Recap: Fourier analysis

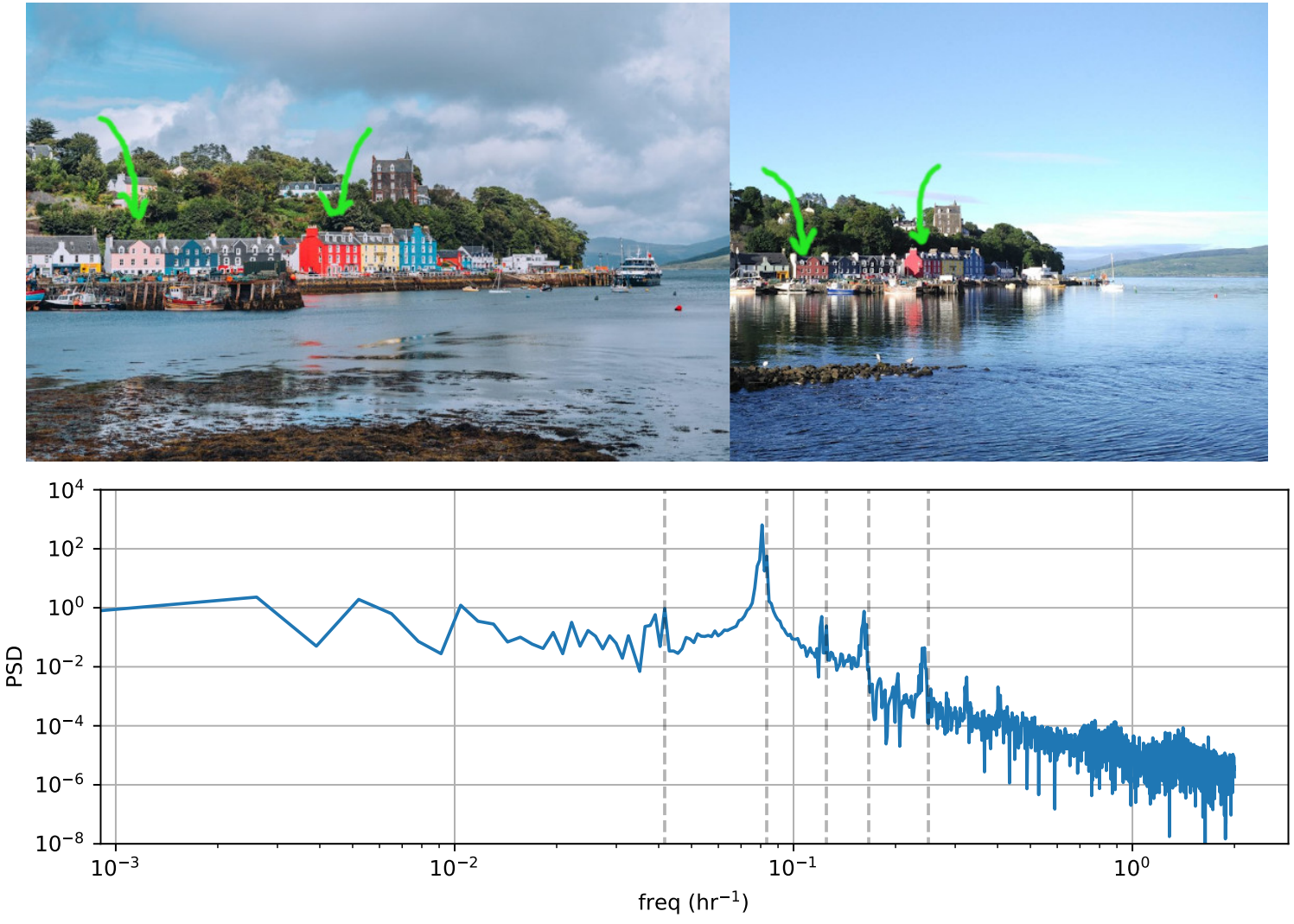

Figure: High and low tide at Tobermory, and its power spectrum computed from tide gauge data.

Recap: Fourier analysis

- can manipulate Fourier coefficients to do various operations
  $\rightarrow$ transform to spectral space, manipulate, transform back to real space

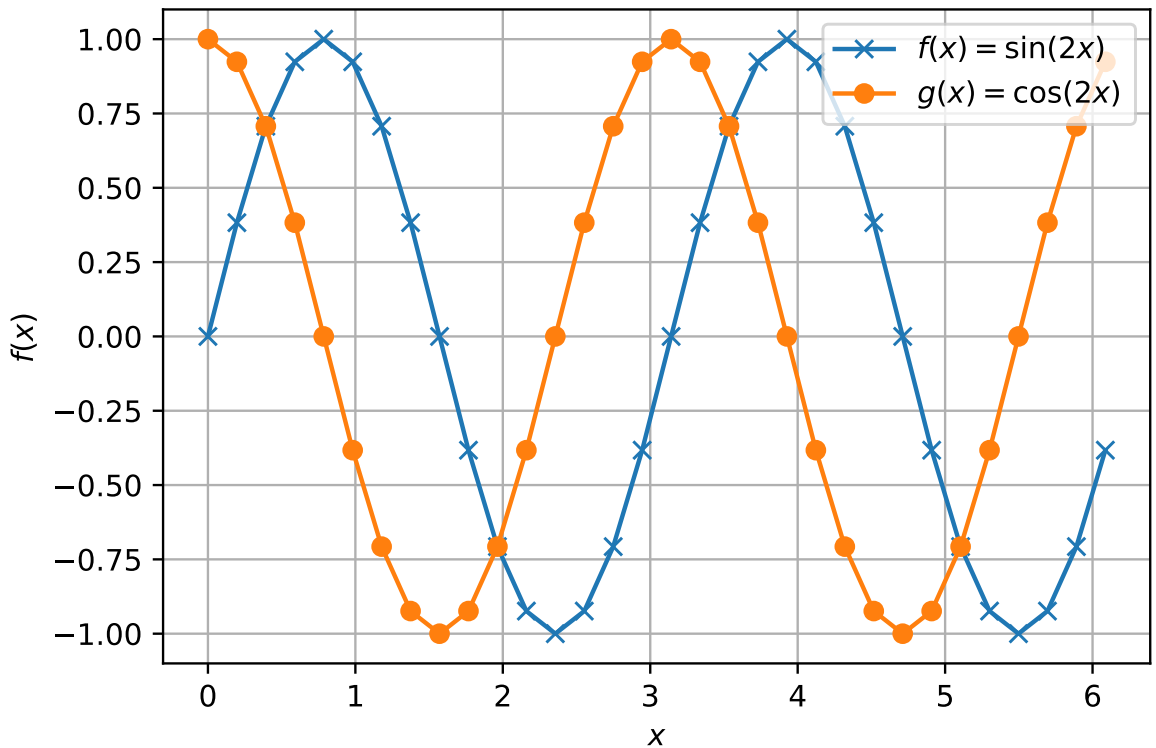

Figure: Example result from manipulating the Fourier coefficients in spectral space: shifting a wave.

Recap: Fourier analysis

- multiply all c_k by ik and $(ik)^2 = -k^2$

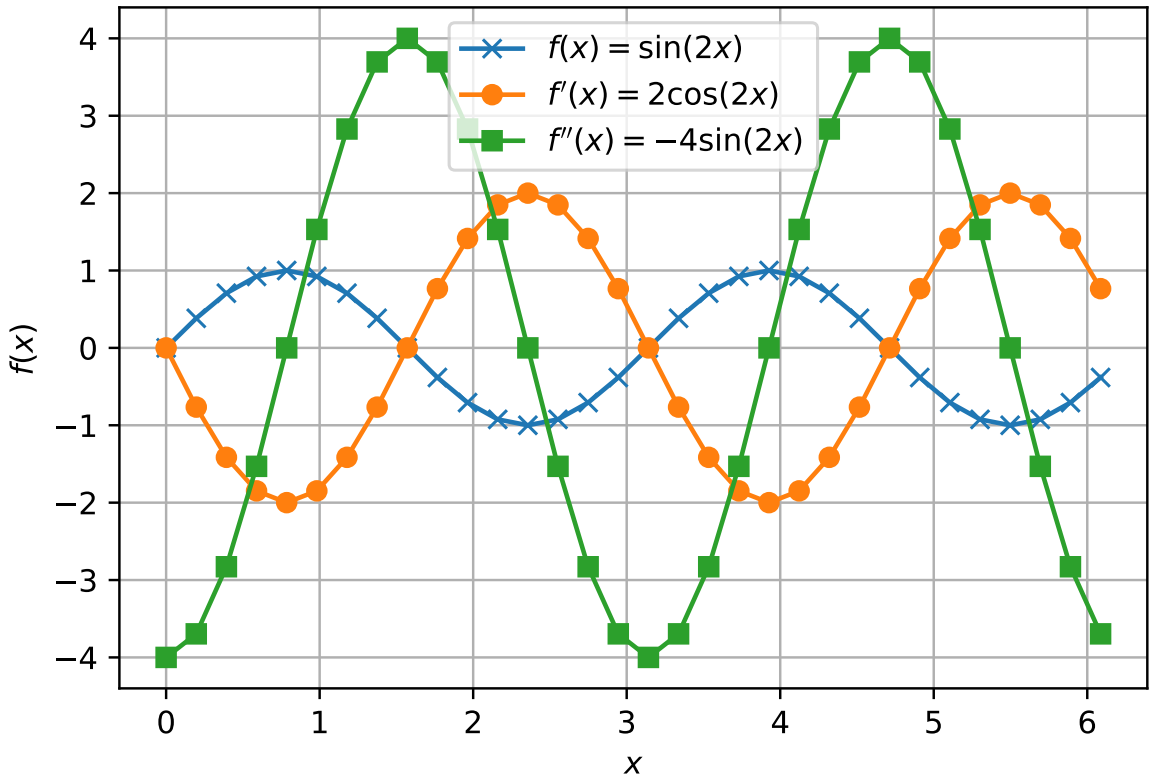

Figure: Example result from manipulating the Fourier coefficients in spectral space: taking derivatives.

Recap: Fourier analysis

- other manipulations possible, but main one of interest comes from the spectral convolution theorem:

convolution in one space

$\equiv$

pointwise multiplication in another

Recap: Fourier analysis

- other manipulations possible, but main one of interest comes from the spectral convolution theorem:

convolution in one space

$\equiv$

pointwise multiplication in another

- can learn convolutions in real space by learning multiplication in spectral space

  $\rightarrow$ FNOs like CNNs but with domain-wide kernel

  $\rightarrow$ can use similar machinery as before, just need to transform the data into spectral space

FNO

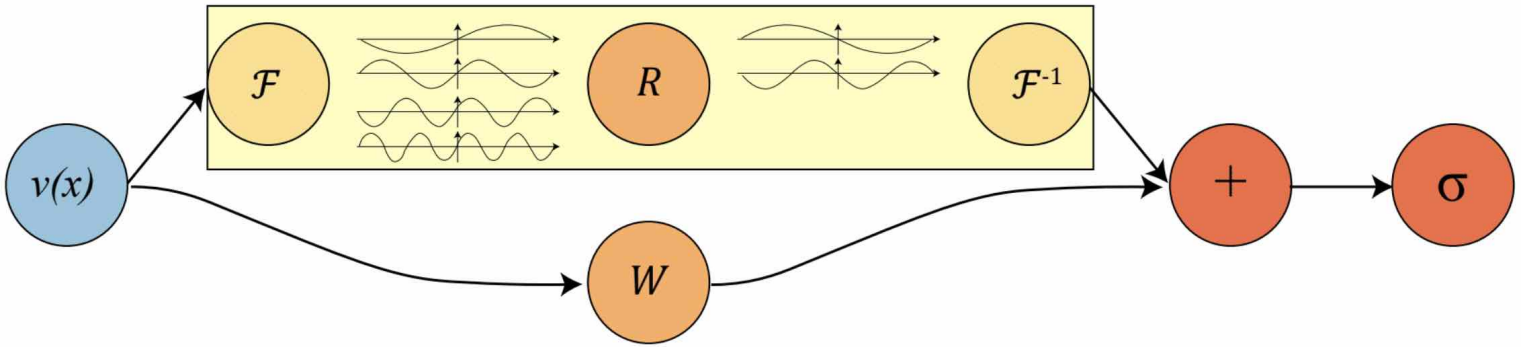

Figure: Schematic for the FNO block. From Fig. 2 of Li et al (2021).

- **lifting layer** $v(x)$: expands the number of channels
- **Fourier layer** $\mathcal{F}^{-1}R\mathcal{F}$: transform ($\mathcal{F}$), learn what to multiply in spectral space ($\mathcal{R}$), transform back ($\mathcal{F}^{-1}$)
- **biases** W: convolution in real space (so multiplication in spectral space) for mode mixing
- add together $+$, then do activation σ

  $\rightarrow$ nonlinear activation function to do some **mode mixing**
- **project** back down at the end

FNO

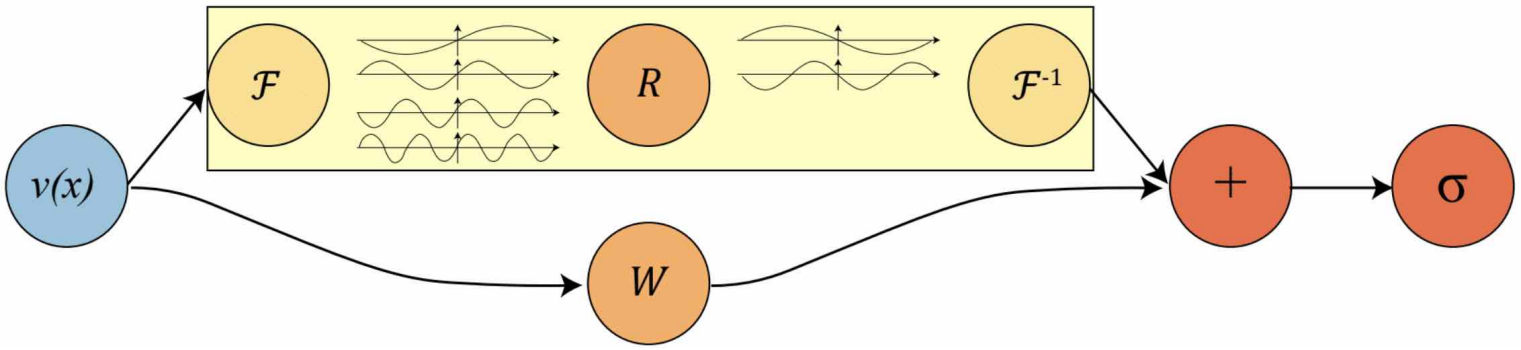

Figure: Schematic for the FNO block. From Fig. 2 of Li et al (2021).

- grid information concatenated into feature space

- non-periodicities?
  $\rightarrow$ can **pad** it (in third example)
- boundary conditions + nonlinearities?
  $\rightarrow$ biases and the activation
- generalisability?
  $\rightarrow$ should be **resolution** independent

FNO1d: linear PDE

- do diffusion equation again

  $\rightarrow$ input is $u(x, t = t_0)$, output is $u(x, t = t_0 + \Delta t)$

  $\rightarrow$ cf. RNN-type testing

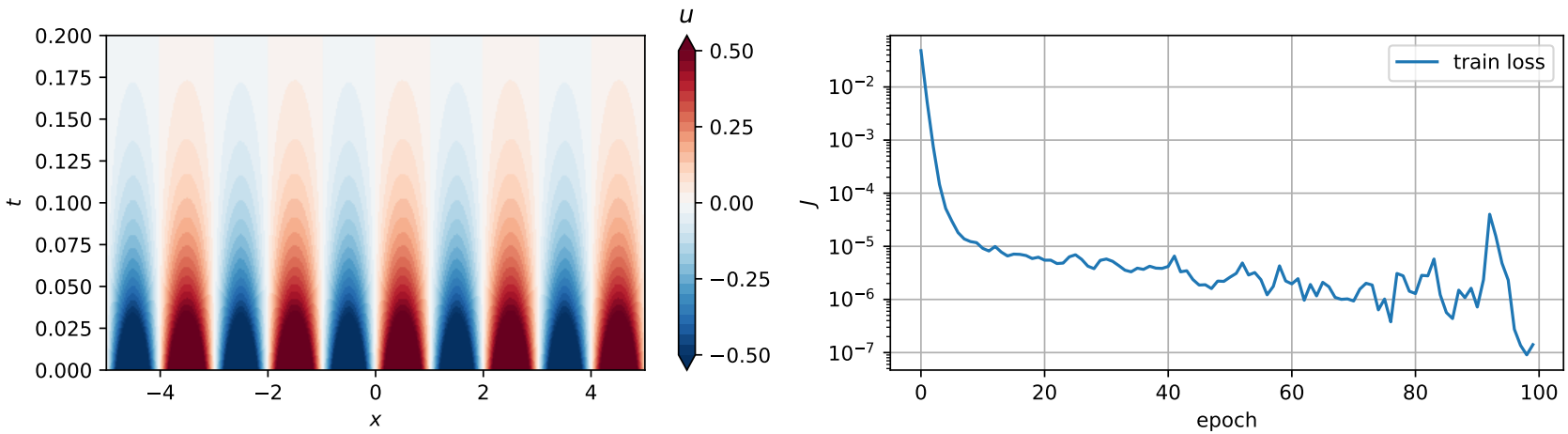

Figure: Data for a particular initialisation and choice of κ from 1d diffusion equation (periodic boundary condition here), and the loss from a simple `FNO1d`.

FNO1d: linear PDE

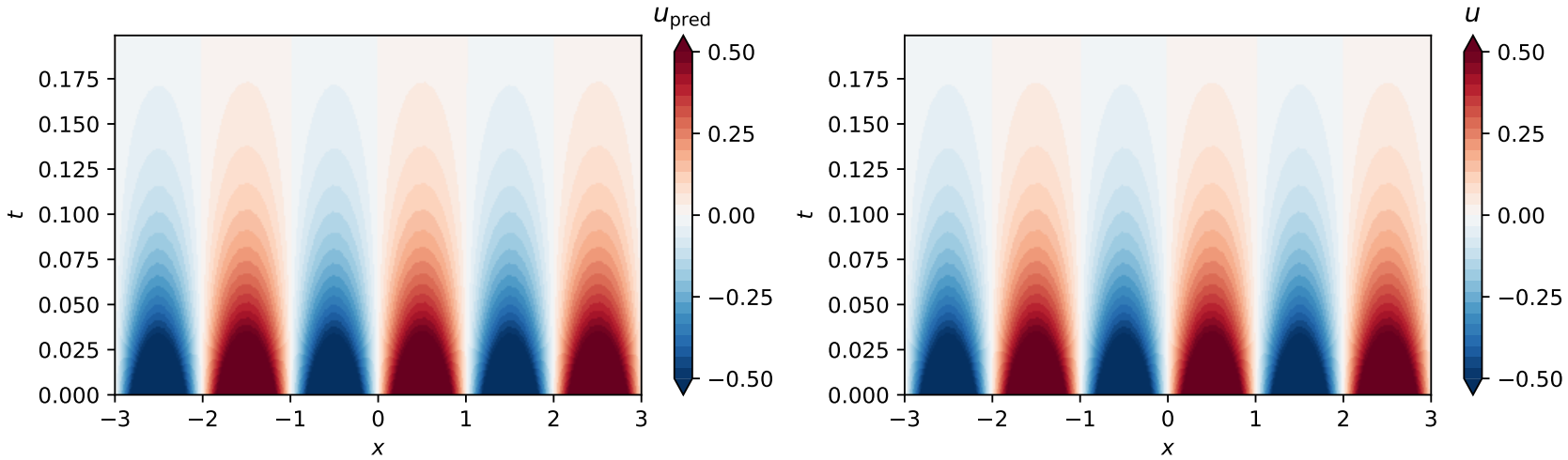

Figure: (Left) One-step prediction: $u_{\text{pred}}(t = t_0 + \Delta t) = F(u(t_0))$. (Right) Model truth.

- one-step prediction: just predict one time-frame from another

  $\rightarrow$ pretty good, as it should be, since that's how we trained it in the first place

FNO1d: linear PDE

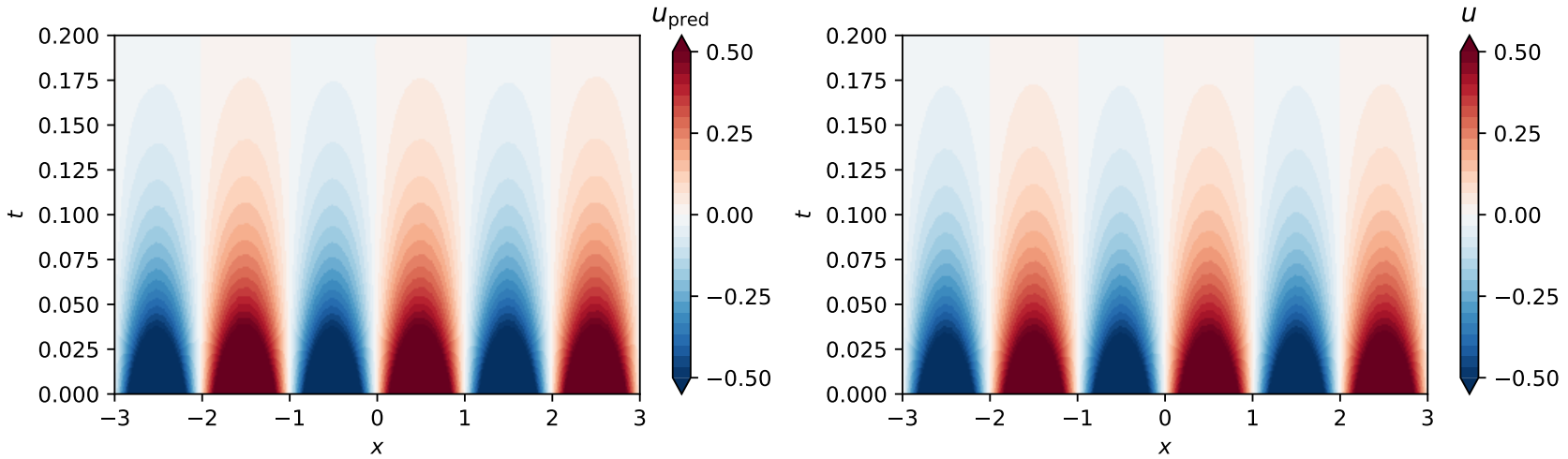

Figure: (Left) Sequential prediction: $u_{\text{pred}}(t = t_0 + n\Delta t) = F^n(u(t_0))$. (Right) Model truth.

- sequential prediction: make one prediction, pass that to model for another prediction and continue

  $\rightarrow$ much harder test

  $\rightarrow$ actually seems ok also!

FNO1d: **non**linear PDE

- do KdV again

  $\rightarrow$ input is $u(x, t = t_0)$, output is $u(x, t = t_0 + \Delta t)$

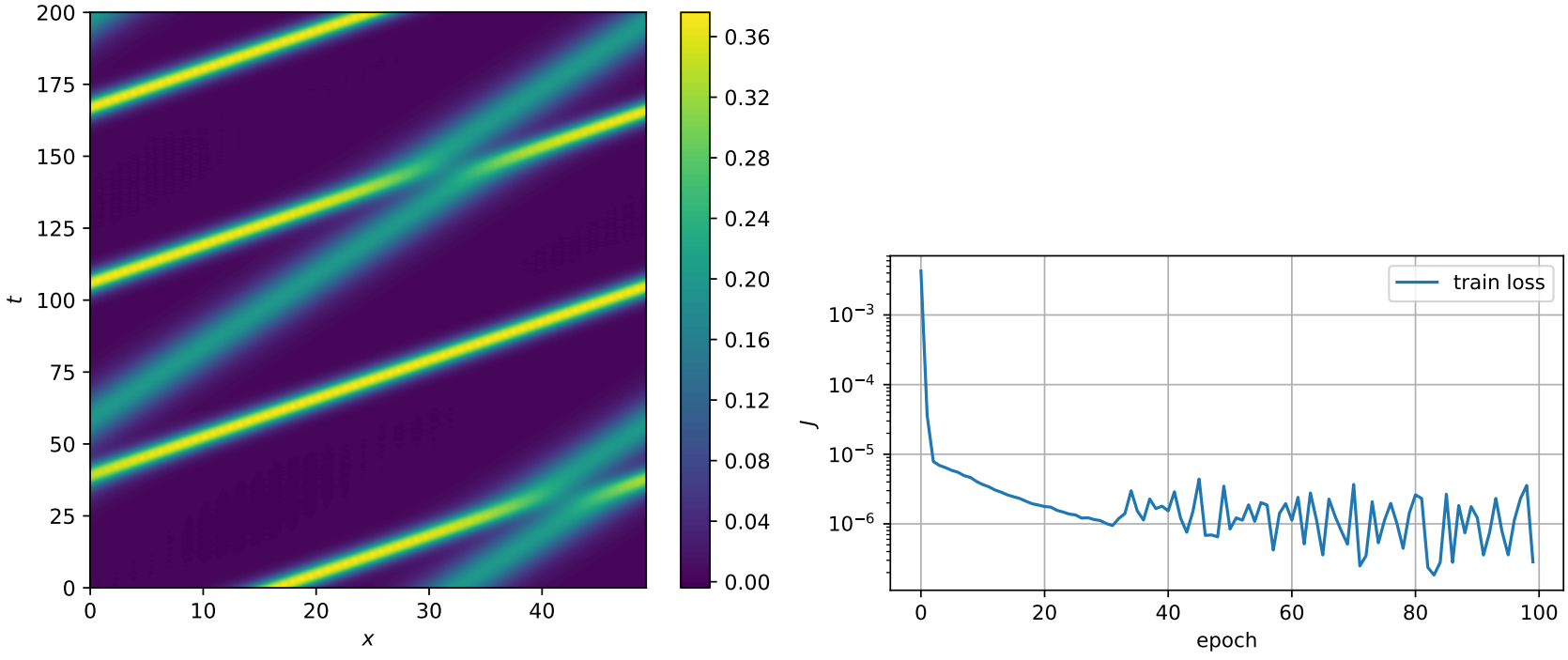

Figure: Data for a particular initialisation from KdV equation (periodic boundary condition here), and the loss from a simple `FNO1d`.

FNO1d: **non**linear PDE

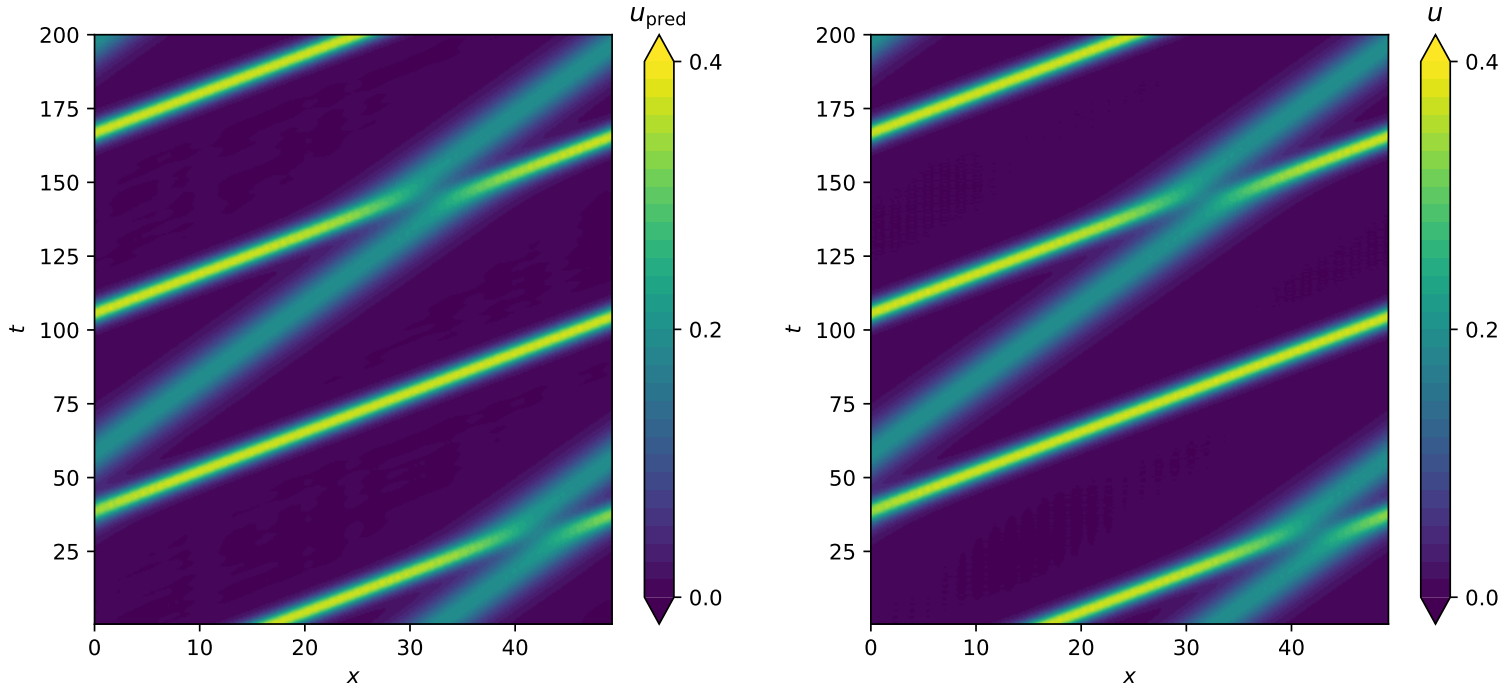

Figure: (Left) One-step prediction: $u_{\text{pred}}(t = t_0 + \Delta t) = F(u(t_0))$. (Right) Model truth.

- one-step prediction: pretty good as it should be

FNO1d: **non**linear PDE

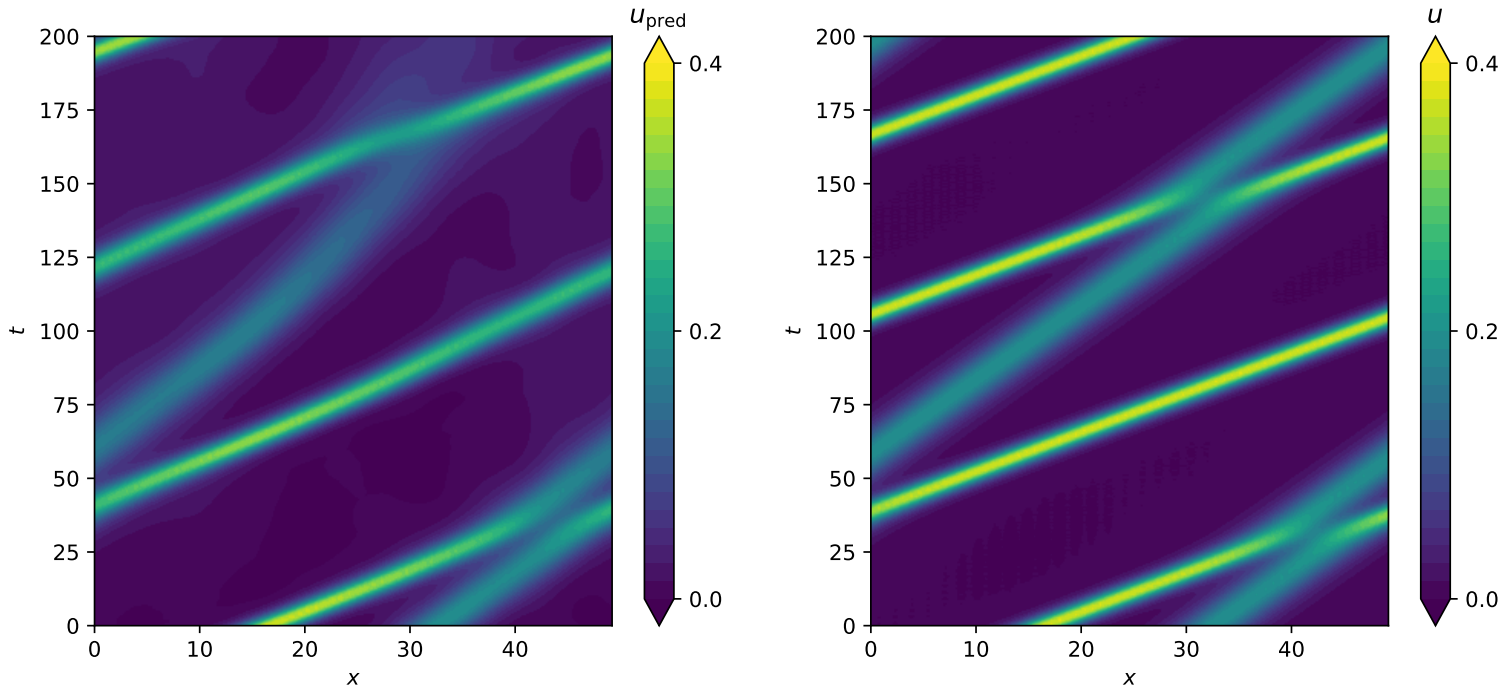

Figure: (Left) Sequential prediction: $u_{\text{pred}}(t = t_0 + n\Delta t) = F^n(u(t_0))$. (Right) Model truth.

- sequential prediction: not so good, but has potential?

FNO2d: image regression

- FNOs restricted to only differential equations and data with periodic boundary conditions?

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- FNOs restricted to only differential equations and data with periodic boundary conditions?

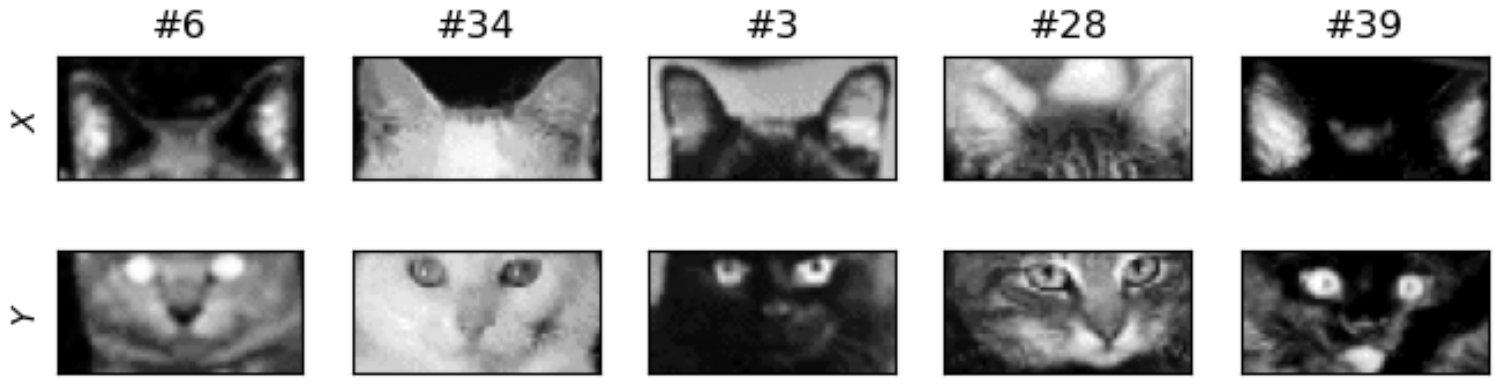


Figure: Unwitting subjects of abuse.

- structure essentially the same, with details to for 2d data

  $\rightarrow$ adds **padding** to data to formally deal with non-periodic data

FNO2d: image regression

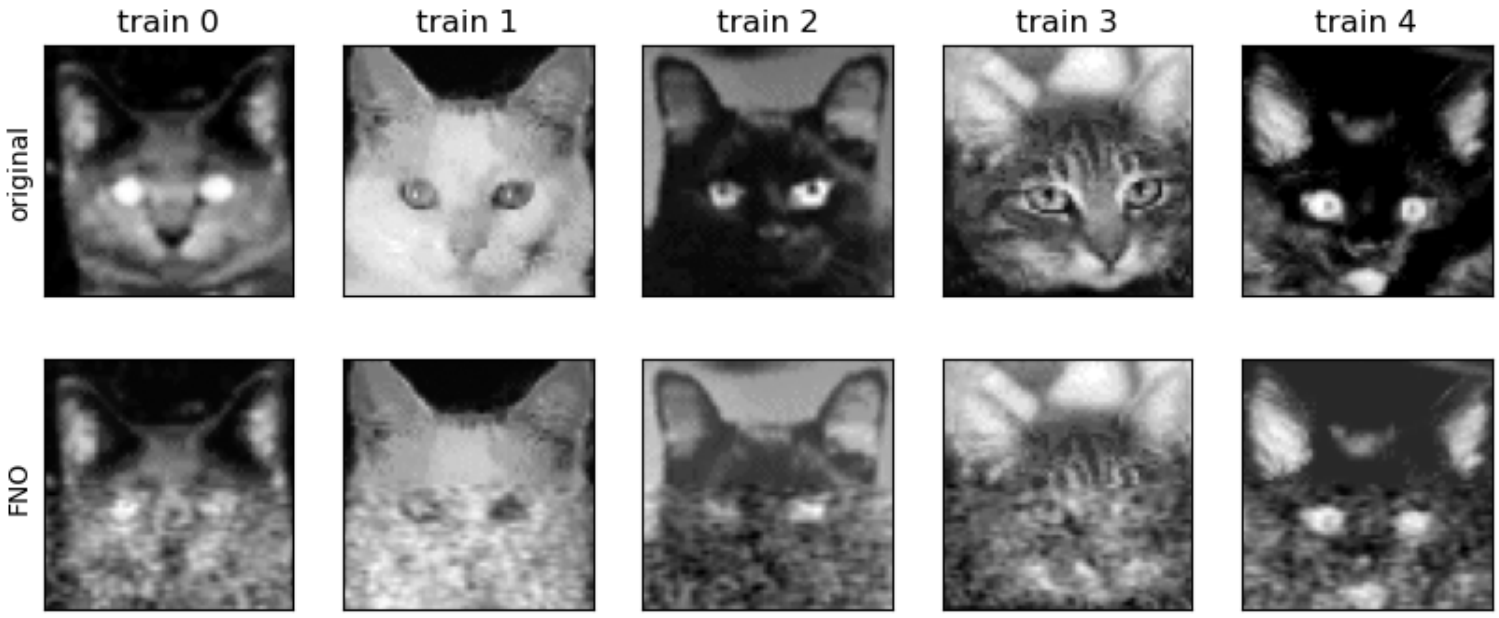


Figure: Unwitting subjects of abuse.

- ...has potential?

FNO2d: image regression

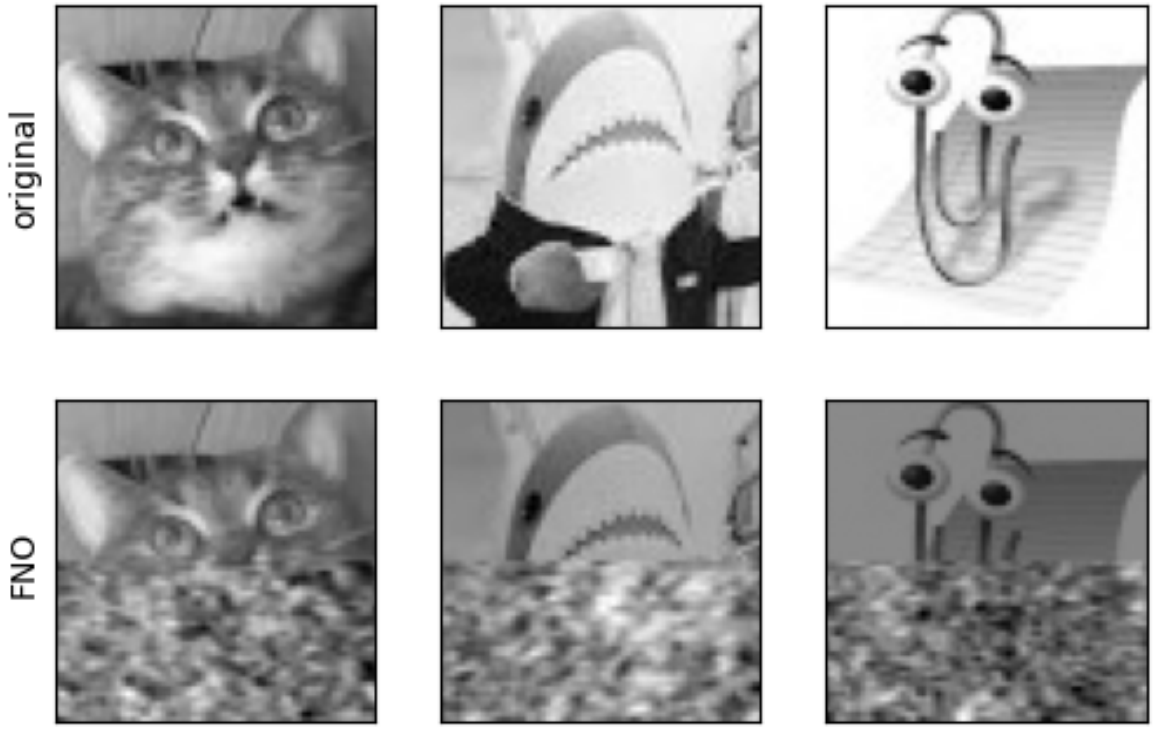


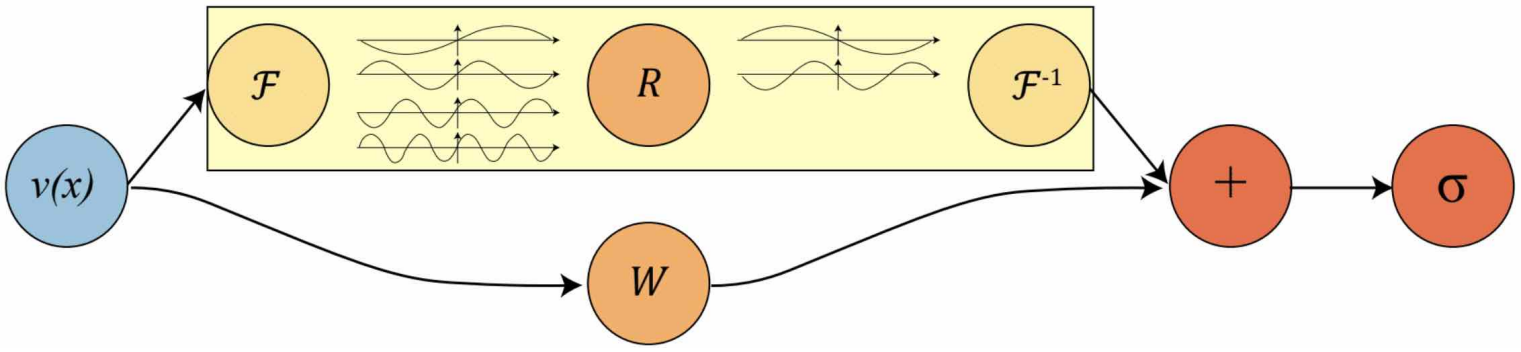
Figure: Further unwitting subjects of abuse.

- just noise here really

  $\rightarrow$ I didn't make the model too complex

  $\rightarrow$ this is a hard problem

Demonstration



- introduced FNOs

  → main keyword is **operator**, the **Fourier** is one realisation

  → probably most suited to differential equations, but can in principle be used for other things too
- variants thereof (e.g. RNOs, PINOs, etc.)

  → there is also the `NeuralOperators` package that I didn't use here (but you should try it)

Example variant: PINO

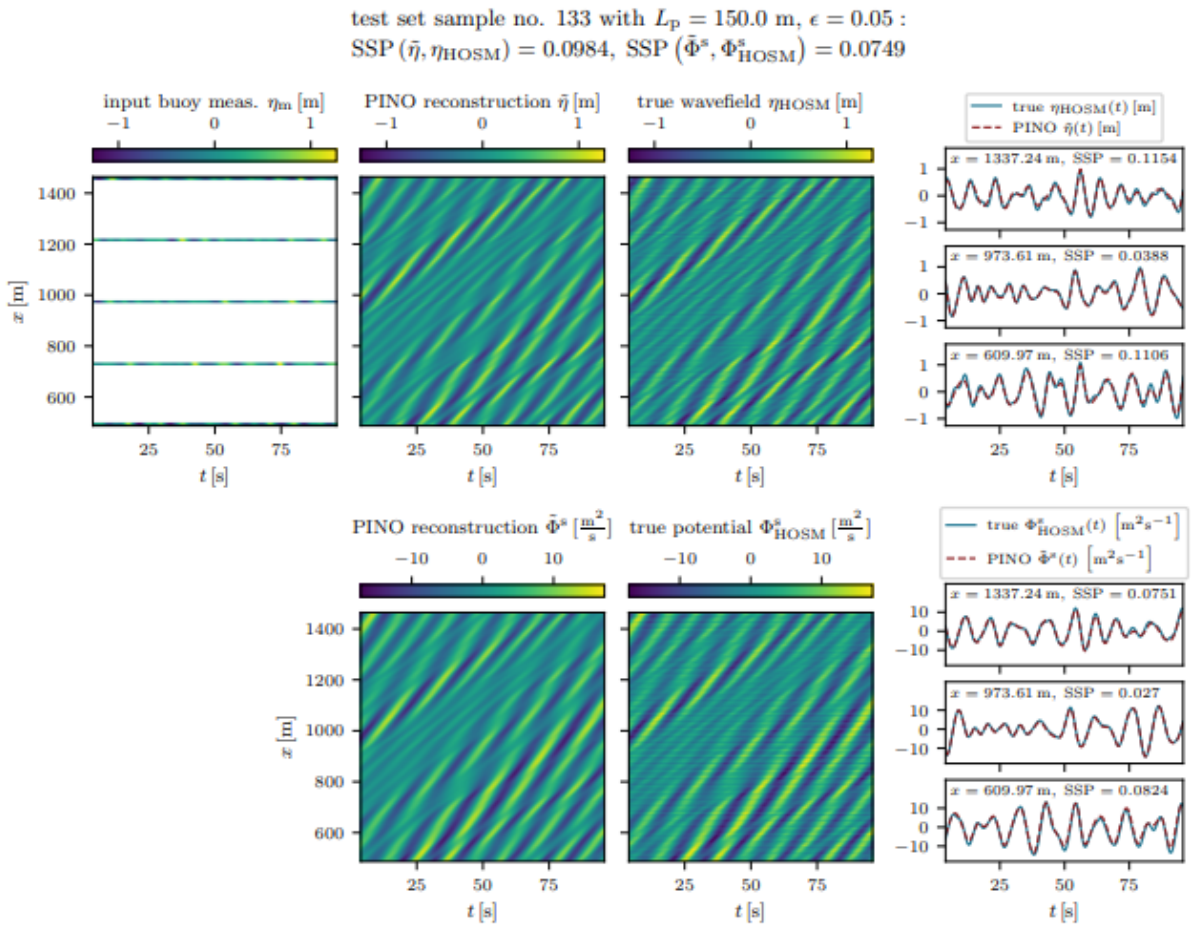


Figure: Reconstructing nonlinear waves with PINO. From Ehlers et al. (2025), Fig. 3.

Moving to a Jupyter notebook →